

Explicit constants for penalty necessity (2D Theorem A), graded meshes, and a 3D note

notes/v6/consts

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Summary

- (1) **Explicit κ_0 (Goal 1) [PROVED; constants computed in exact rational arithmetic, final assembly in floating point with rounding far below the stated digits]**. Theorem 3: let θ be the minimum angle of $\mathcal{T}_h$, e any edge of Γ_h with midpoint x_0 , $0 < Y \leq 10^{-6}$ (and $4Y < L - 1$), $0 < \kappa \leq \kappa_0(\theta)$, and suppose every triangle meeting $B(x_0, 3Y)$ has diameter $\leq \kappa Y$. Then N_h is indefinite on Z_h for every $k \geq 4$ whenever $\mu < 15 h/Y$.

θ	10°	15°	20°	30°	45°	60°
$\kappa_0(\theta)$	0.00379	0.00594	0.00803	0.0117	0.0143	0.0161

Consequently (Cor. 4), if all triangles meeting $B(x_0, 3C_q|e|/\kappa_0)$ have diameter $\leq C_q|e|$, then $\mu^* \geq (15\kappa_0/C_q) h/|e| \geq (15\kappa_0 c_0/C_q) \rho$; e.g. $\mu^* \geq 0.175 c_0 \rho/C_q$ for $\theta \geq 30^\circ$. The witness is the curl of the Argyris interpolant; the proof uses no inf-sup constant and no compactness. The only analytic inputs are Taylor's formula, the exact reference Argyris basis, and elementary triangle geometry.

- (2) **Realistic κ [NUMERICAL]**. The rigorous interpolation bound is pessimistic by a factor $\approx 7 \cdot 10^4$ (checked on 600 random triangles). On actual meshes the same witness already gives $Y(2B - A)/C \geq 15$ at $\kappa = \text{diam } T/Y \approx 0.85$ (uniform) to 1.8 (graded), against the proved $\kappa_0 \approx 0.01$.

- (3) **Graded meshes (Goal 2).**

- **[PROVED]** The quasi-uniform region may sit at *any* boundary edge, not only the shortest one; (M0) pays only a factor c_0 .
- **[PROVED, obstruction]** The proposed geometric-growth argument cannot remove the hypothesis. Shape regularity only gives the *upper* bound $\text{diam } T \leq C(\theta)(|e| + \text{dist}(T, x_0))$ with $C(\theta) \geq 1$, and that growth is attained. On a dyadically graded boundary layer (min angle 26.57° , (M0)–(M2)) *every* ball $B(x_0, 3Y)$, $x_0 \in \Gamma_h$, $Y > 0$, contains a triangle of diameter $\geq Y$. So the hypothesis fails for every $\kappa < 1$ at every scale, and Theorem 3 (any version with $\kappa_0 < 1$) never applies there.
- **[NUMERICAL]** Nevertheless the *same witness* works on such meshes. On the dyadic layer (ratio 2 per layer) and a triadic one (ratio 3, min angle 15.3°), $Y(2B - A)/C = 15.28$ and 15.24 at $Y = 2h_\Gamma$, and $\rightarrow 15.487$ as Y/h_Γ grows. So on those meshes $\mu^* \geq 7.6 h/h_\Gamma$. The obstruction is to the proof (interpolation constants on coarse cells), not to the conclusion.
- Status: general (M0)–(M2) remains **[OPEN]**, exactly as in the penalty report (Theorem B there). The explicit theorem needs about $3C_q/\kappa_0 \approx 250$ –800 quasi-uniform layers; the numerics suggest 2–3.

- (4) **3D (Goal 3) [NOT DONE]**. Why the 2D route does not transfer is explained in §6: Theorem 3B's constant runs through $\beta(\text{IS3})$, itself non-explicit ([GN18]); and `td_k3b`'s κ_0 needs a quantitative Star Nondegeneracy Theorem.

1 Setting and notation

As in `02_setting.tex`: $\Omega_h = R \setminus P_h$, P_h a convex polygon inscribed in the unit circle, $\mathcal{T}_h$ conforming on Ω_h , $k \geq 4$, $Z_h = \{v \in V_h^R : \operatorname{div} v = 0\}$. For $v \in Z_h$ put $A(v) = \int_{\Omega_h} \frac{1}{2} Dv : Dv$ with $Dv = \nabla v + \nabla v^\top$, $B(v) = \langle \partial_n v, v \rangle_{\Gamma_h}$ with $\partial_n v = (\nabla v)n$ and n the outward normal of the fluid, and $C(v) = \|v\|_{\Gamma_h}^2$. Then $N_h(v, v) = A - 2B + (\mu/h)C$, and A, B are dilation invariant while C scales linearly. So

$$N_h(v, v) = \frac{C(v)}{Y} \left[\frac{\mu Y}{h} - \mathcal{Q}(v) \right], \quad \mathcal{Q}(v) := \frac{Y(2B(v) - A(v))}{C(v)}.$$

θ denotes the minimum angle of $\mathcal{T}_h$. For a symmetric j -linear form S , $|S|$ is the operator norm, which equals $\sup_{|u|=1} |S[u^j]|$ (Banach). $|\cdot|_F$ is the Frobenius norm.

Profile. $g(t) = t(1-t)^6$ for $-\frac{1}{8} \leq t \leq 1$, $g = 0$ for $t \geq 1$; $\phi(s) = (1-s^2)^6$ on $[-1, 1]$, 0 outside. Both are C^5 with piecewise-polynomial, bounded sixth derivative. With $X = 2Y$,

$$F(x, y) = Y \phi(x/X) g(y/Y), \quad w = \operatorname{curl} F = (F_y, -F_x),$$

in coordinates in which e lies on $\{y = 0\}$, $x_0 = 0$, and the fluid is locally $\{y > 0\}$. On the part of Ω_h below $y = -Y/8$ (the far side of P_h) set $F := 0$. That part and the part near x_0 are separated by P_h , so no triangle meets both (Lemma 2).

Scaled derivative bounds (`argyris_consts.py`, exact critical points). Let $m_j := Y^{j-1} \sup |D^j F|$ over $\{y \geq -Y/8\}$. Bounding each partial derivative termwise, with $\max_{|u|=1} |u_1|^i |u_2|^{j-i}$ in place of the monomials,

$$m_1 \leq 3.6635, \quad m_2 \leq 32.187, \quad m_3 \leq 214.88, \quad m_6 \leq 18918.2.$$

Half-plane values (exact, SymPy; $Y = 1$): $\hat{A} = \frac{4717730594816}{558289907175} = 8.45032$, $\hat{B} = \frac{201326592}{16900975} = 11.91213$, $\hat{C} = \frac{16777216}{16900975} = 0.992677$. So $\hat{\mathcal{Q}} = (2\hat{B} - \hat{A})/\hat{C} = 15.48734$, which agrees with the penalty report's 15.487.

2 Explicit Argyris interpolation bounds

Lemma 1 (Explicit local bound). *[PROVED] Let T be a triangle with minimum angle $\geq \theta$ and diameter h_T , and let $f \in C^5(T)$ have $D^5 f$ Lipschitz, with $\operatorname{ess\,sup}_T |D^6 f| \leq M_6$. Let I be the Argyris (C^1 - P_5) interpolant. Its DOFs are the values, gradients and Hessians at the vertices and the unit-normal derivatives at the edge midpoints. Then on T , for $j = 1, 2$,*

$$|D^j(f - If)| \leq E_j(\theta) M_6 h_T^{6-j},$$

where

$$E_j(\theta) = \frac{1}{(6-j)!} + \left(\frac{\sqrt{2}}{\sin \theta s(\theta)} \right)^j \left[P_j + \left(\frac{1}{60} + 2c_H \right) N_{0,j} + \left(\frac{1}{3840} + \frac{\sqrt{2}c_H}{2 \sin \theta} \right) (N_{1,j} + N_{2,j}) \right].$$

Here $s(\theta) = \min(\sqrt{3}/2, \sin 2\theta)$ and $c_H = \frac{29}{1920}$. The reference constants are

$$P_1 \leq 0.19597, P_2 \leq 1.40308, \quad (N_{0,1}, N_{1,1}, N_{2,1}) \leq (1.8857, 3.7713, 3.7713), \quad (N_{0,2}, N_{1,2}, N_{2,2}) \leq (16.866, 33.731)$$

Numerically: $E_1 = 17.9, 6.54, 3.37, 1.47, 0.933, 0.724$ and $E_2 = 3606, 597, 179, 39.0, 17.3, 10.9$ for $\theta = 10^\circ, 15^\circ, 20^\circ, 30^\circ, 45^\circ, 60^\circ$.

Proof. 1. *Taylor.* Let a_0 be the vertex with the largest angle $\alpha \in [\pi/3, \pi - 2\theta]$, so $\sin \alpha \geq s(\theta)$. Let p be the degree-5 Taylor polynomial of f at a_0 and $r = f - p$. Since $Ip = p$, $f - If = r - Ir$. Along each segment $[a_0, x] \subset T$, $D^5 f$ is Lipschitz, so the integral remainder gives

$$|D^m r(x)| \leq M_6 |x - a_0|^{6-m} / (6-m)! \leq M_6 h_T^{6-m} / (6-m)!,$$

and all derivatives of r of order ≤ 5 vanish at a_0 .

2. *Reference map.* Let $\Phi(\hat{x}) = J\hat{x} + a_0$ map $\hat{T} = \text{conv}\{(0,0), (1,0), (0,1)\}$ onto T . The columns of J are the edges E_1, E_2 at a_0 , of lengths ℓ_1, ℓ_2 . By the law of sines, $h_T \sin \theta \leq \ell_i \leq h_T$, and the longest edge E_0 (opposite a_0) has length h_T . Hence

$$\sigma_{\max}(J) \leq \sqrt{2}h_T, \quad \|J^{-1}\| = \sigma_{\max}/\det J \leq \frac{\sqrt{\ell_1^{-2} + \ell_2^{-2}}}{\sin \alpha} \leq \frac{\sqrt{2}}{h_T \sin \theta s(\theta)}.$$

3. *Pulled-back interpolant.* Let $\hat{q} = (Ir) \circ \Phi \in P_5$. Its vertex values, gradients and Hessians are those of $\hat{r} = r \circ \Phi$, because these DOF sets are affine-equivalent. Only the edge DOFs differ. On reference edge $\hat{e}$ with edge vector $\hat{t}$ (the reference DOF being $\nabla \hat{q}(\hat{m}) \cdot \hat{\nu}$, with $\hat{\nu} \in \{(1,1), (-1,0), (0,-1)\}$), write $J^{-1}n_E = a\hat{\nu} + b\hat{t}$. Then $\gamma_E := \nabla \hat{q}(\hat{m}) \cdot \hat{\nu} = [n_E \cdot \nabla r(m_E) - b(Hf_E)'(\frac{1}{2})]/a$. Here $f_E(s) = \hat{r}(\hat{p} + s\hat{t})$, and Hf_E is its quintic Hermite interpolant from f, f', f'' at $s = 0, 1$, which is the restriction of $\hat{q}$. Exactly, $(Hf)'(\frac{1}{2}) = \frac{15}{8}(f_1 - f_0) - \frac{7}{16}(f'_0 + f'_1) + \frac{1}{32}(f''_1 - f''_0)$.

4. *Coefficient bounds.* Since $\det(J^{-1}n_E, \hat{t}) = \pm \ell_E / \det J$,

$$|a|^{-1} = \det J |\det(\hat{\nu}, \hat{t})| / \ell_E, \quad |b/a| \leq |\hat{\nu}| \sigma_{\max}(J) / \ell_E.$$

This gives:

- on the hypotenuse (E_0): $|a|^{-1} \leq 2h_T$ and $|b/a| \leq 2$;
- on the legs: $|a|^{-1} \leq h_T$ and $|b/a| \leq \sqrt{2}/\sin \theta$.

With $f_E^{(m)} = D^m r[E^m]$ and step 1:

- $|\gamma_{E_0}| \leq M_6 h_T^6 (\frac{2}{120} + 2c_H)$;
- $|\gamma_{E_{1,2}}| \leq M_6 h_T^6 (\frac{2^{-5}}{120} + \frac{\sqrt{2}}{\sin \theta} \frac{c_H}{2})$, since the data at a_0 vanish.

The vertex DOFs at a_1, a_2 are $D^m r[E_{i_1}, \dots, E_{i_m}]$, of size $\leq M_6 h_T^6 / (6-m)!$.

5. *Reference basis.* Expand $\hat{q}$ in the exact reference Argyris basis (21×21 inverse over $\mathbb{Q}$). Bound each basis function's partial derivatives of order j on $\hat{T}$ by the largest absolute Bernstein coefficient, which is a rigorous sup bound. Combine them into operator-norm bounds via the Frobenius norm. This gives $|\hat{D}^j \hat{q}| \leq M_6 h_T^6 [P_j + \dots]$, the bracket in E_j .

6. *Back to T .* $|D^j(Ir)| \leq \|J^{-1}\|^j |\hat{D}^j \hat{q}|$. Add $|D^j r|$ from step 1. $\square$

Check (validate_interp.py, [NUMERICAL]). Physical Argyris interpolation of F on 600 random triangles ($\theta \in [10^\circ, 53^\circ]$, $h_T \in [0.01, 0.2]$, random positions in $\text{supp } F$): the ratio of actual error to the bound is at most $1.8 \cdot 10^{-5}$ ($j = 1$) and $1.4 \cdot 10^{-5}$ ($j = 2$). The bound holds, and is very loose.

3 Explicit Theorem A

Lemma 2 (Local geometry of Γ_h). *[PROVED] Suppose every boundary edge meeting $B(x_0, 3Y)$ has length $\leq \kappa Y$, with $\kappa \leq 1$ and $Y \leq 10^{-2}$. Put $X' = (2 + \kappa)Y$, $\omega_X = \arcsin(X'/(1 - \kappa^2 Y^2/4))$,*

$$\delta = \frac{1}{4}\kappa^2 Y^2 + \frac{1}{2}\omega_X^2, \quad \sigma = \tan(\omega_X + \arcsin(\kappa Y/2)).$$

Then on $|x| \leq X'$ the part of Γ_h near x_0 is a graph $y = \gamma(x)$ with $-\delta \leq \gamma \leq 0$ and $|\gamma'| \leq \sigma$, and $\Omega_h \cap \{|x| \leq X', y > -Y/8\} = \{y > \gamma(x)\}$ (this uses $\delta \leq Y/8$, true for $Y \leq 10^{-2}$).

Proof. • $\gamma \leq 0$. P_h is convex and e is one of its edges, so P_h lies in $\{y \leq 0\}$.

- *Polar description.* Let $O = (0, -d)$ be the centre. A boundary point on a chord of length $\ell' \leq \kappa Y$ is $O + r(\sin \omega, \cos \omega)$, with $r \geq \sqrt{1 - \ell'^2/4} \geq 1 - \ell'^2/4$.
- *Depth.* So $y \geq -\ell'^2/4 - (1 - \cos \omega)$, while $|x| \leq X'$ forces $|\omega| \leq \omega_X$.
- *Slope.* A chord between polar angles ω_1, ω_2 has direction angle $(\omega_1 + \omega_2)/2$, and $|\omega_2 - \omega_1| \leq 2 \arcsin(\ell'/2)$.
- *Fluid side.* The region between γ and $y = -Y/8$ lies in P_h by convexity. $\square$

Theorem 3 (Theorem A with explicit constants). *[PROVED] Let $e \in \mathcal{E}_\Gamma$ have midpoint x_0 , let $0 < Y \leq Y_0 := 10^{-6}$ with $4Y < L - 1$, let $0 < \kappa \leq \kappa_0(\theta)$ (table in the Summary), and suppose every $T \in \mathcal{T}_h$ meeting $B(x_0, 3Y)$ has $\text{diam } T \leq \kappa Y$. Let $v := \text{curl } IF$, with I the global Argyris interpolant. Then $v \in Z_h$ for every $k \geq 4$ and $\mathcal{Q}(v) \geq 15$. Hence N_h is indefinite on Z_h whenever $\mu < 15h/Y$.*

Proof. 1. *Membership in Z_h .*

- IF is globally C^1 and piecewise P_5 , so v is continuous, piecewise $P_4 \subset P_k$, and pointwise divergence-free.
- $IF|_T = 0$ on every T not meeting $\text{supp } F \cap \overline{\Omega_h} \subset \{|x| \leq 2Y, -\delta \leq y \leq Y\} \subset B(x_0, \sqrt{5}Y + \delta)$.
- So $\text{supp } v \subset S := \bigcup \{T : T \cap \text{supp } F \neq \emptyset\} \subset \{|x| \leq X', \gamma \leq y \leq (1 + \kappa)Y\}$, which misses ∂R .
- Every $T \subset S$ lies in $\{y \geq -\delta\}$, where F is given by the profile formula, so Lemma 1 applies with $M_6 = m_6 Y^{-5}$ and $h_T \leq \kappa Y$.

Write $e_F = F - IF$, $|S| \leq 2X'((1 + \kappa)Y + \delta)$ and $\Lambda := |\Gamma_h \cap S| \leq 2X'\sqrt{1 + \sigma^2}$.

2. *Continuum field on the true geometry.* By Lemma 2, $\Omega_h \supset \{0 < y < Y, |x| < X\}$, so $A_{\Omega_h}(w) = \hat{A} + A_{\text{sl}}$ with $0 \leq A_{\text{sl}} \leq 4 \sup |D^2 F|^2 \cdot 2X\delta = 16m_2^2\delta/Y$.

- *Boundary term.* Using $n ds = (\gamma', -1) dx$ and moving from $y = 0$ to $y = \gamma$: $|B_{\Gamma_h}(w) - \hat{B}| \leq \varepsilon_B := 4[\sigma m_1 m_2 + (\delta/Y)(m_1 m_3 + m_2^2)]$.
- *Trace term.* $|C_{\Gamma_h}(w) - Y\hat{C}| \leq \varepsilon_C := 4Y[m_1^2(\sqrt{1 + \sigma^2} - 1) + 2\sqrt{1 + \sigma^2} m_1 m_2 \delta/Y]$.

3. *Interpolation.* $v - w = \text{curl } e_F$, so $|\nabla(v - w)|_F = |D^2 e_F|_F \leq \sqrt{2}|D^2 e_F|$ and $|v - w| = |\nabla e_F|$. By Lemma 1, $|D^2 e_F| \leq E_2 m_6 \kappa^4/Y$ and $|\nabla e_F| \leq E_1 m_6 \kappa^5$ on S . With $A(z)^{1/2} \leq \sqrt{2}\|\nabla z\|$:

$$A(v)^{1/2} \leq (\hat{A} + A_{\text{sl}})^{1/2} + \eta_A, \quad \eta_A = 2E_2 m_6 \kappa^4 \sqrt{|S|}/Y; \quad C(v)^{1/2} \leq (Y\hat{C} + \varepsilon_C)^{1/2} + \eta_C, \quad \eta_C = E_1 m_6 \kappa^5 \sqrt{\Lambda};$$

$$|B(v) - B_{\Gamma_h}(w)| \leq \varepsilon'_B := \frac{E_2 m_6 \kappa^4 \sqrt{\Lambda}}{Y} ((Y\hat{C} + \varepsilon_C)^{1/2} + \eta_C) + \frac{m_2 \sqrt{\Lambda}}{Y} \eta_C,$$

using $B(v) - B(w) = \langle \partial_n(v - w), v \rangle + \langle \partial_n w, v - w \rangle$.

4. Assembly.

$$\mathcal{Q}(v) \geq \mathcal{Q}_{\text{lb}} := \frac{Y[2(\hat{B} - \varepsilon_B - \varepsilon'_B) - ((\hat{A} + A_{\text{sl}})^{1/2} + \eta_A)^2]}{((Y\hat{C} + \varepsilon_C)^{1/2} + \eta_C)^2}.$$

Every error term is nondecreasing in κ and in Y . $\mathcal{Q}_{\text{lb}}$ is dimensionless: $|S|/Y^2$, Λ/Y , δ/Y and σ depend only on κ and Y through the formulas above. The table value $\kappa_0(\theta)$ is the bisection root of $\mathcal{Q}_{\text{lb}}(\kappa, \theta, Y_0) = 15$, rounded down (`assemble.py`, `assemble.log`). A grid check below (κ_0, Y_0) confirms $\min \mathcal{Q}_{\text{lb}} \geq 15.0000$. As $\kappa, Y \rightarrow 0$, $\mathcal{Q}_{\text{lb}} \rightarrow 15.48734$. $\square$

Corollary 4 (ρ -scaling). *[PROVED] Assume (M0). Suppose that for some edge $e \in \mathcal{E}_\Gamma$, with midpoint x_0 , every triangle meeting $B(x_0, 3C_q|e|/\kappa_0)$ has diameter $\leq C_q|e|$, where $C_q \geq 1$ and $C_q|e|/\kappa_0 \leq Y_0$. Then*

$$\mu^* \geq \frac{15\kappa_0(\theta)}{C_q} \frac{h}{|e|} \geq \frac{15c_0\kappa_0(\theta)}{C_q} \rho.$$

Proof. Take $Y = C_q|e|/\kappa_0$ in Theorem 3. For the second inequality, $|e| \leq h_\Gamma \leq \min |e'|/c_0$ by (M0). $\square$

Compared with the penalty report:

- κ_0 is explicit;
- the edge need not be the shortest one;
- the constant 15 is kept, not degraded to $15 - \varepsilon(\kappa_0)$;
- the curvature and sliver terms flagged by referee 1 are handled explicitly (Lemma 2, step 2).

$Y_0 = 10^{-6}$ is a crude, fully explicit smallness condition (at $Y_0 = 3 \cdot 10^{-6}$ the values of κ_0 drop by about 15%, see the log; for $Y \gtrsim 5 \cdot 10^{-6}$, $\mathcal{Q}_{\text{lb}} < 15$ even as $\kappa \rightarrow 0$). It comes almost entirely from A_{sl} and ε_B , which are $O(Y)$ with large profile constants.

Remark 5 (Scope). Theorem 3 covers $k \geq 4$ (Argyris). For Clough–Tocher, $k \geq 2$, the same proof applies with the HCT reference basis, but I did not compute those constants.

4 How far the explicit κ_0 is from the truth

[NUMERICAL] (`realistic_kappa.py`, `.log`). On half-plane meshes (squares of side s split by a diagonal, optionally with random vertex perturbation $\pm s/4$), $\mathcal{Q}(\text{curl } IF)$ with $Y = 1$ is:

$\kappa = \text{diam } T/Y$	1.41	1.13	0.85	0.71	0.42	0.14	0.07
uniform (45°)	12.15	14.06	15.06	15.28	15.46	15.487	15.487

The perturbed meshes (min angle 9–27°) give 14.79 at $\kappa = 1.06$ and 15.28 at $\kappa = 0.88$. So the true threshold is $\kappa \approx 0.85$ –1, against the proved 0.004–0.016. The loss comes from the Taylor/sup-norm bound ($\approx 7 \cdot 10^4$ in E_j , entering as a fourth root) and from the $\sin \theta$ factors.

5 Graded meshes (Goal 2)

Why geometric growth does not help [PROVED].

- *What shape regularity gives.* Triangles sharing an edge have diameter ratio $\leq 1/\sin \theta$, and a vertex has at most $2\pi/\theta$ triangles. From these one gets only $\text{diam } T \leq C(\theta)(|e| + \text{dist}(T, x_0))$, with $C(\theta) \geq 1$.

- *What the theorem needs.* Every triangle in $B(x_0, 3Y)$ must have $\text{diam } T \leq \kappa_0 Y$, with $\kappa_0 \ll 1$ (and, numerically, $\lesssim 1$ for any version of this witness).
- *Why they do not meet.* The upper bound gives $\text{diam} \leq C(\theta)(|e| + 3Y) \geq 3Y$ at the rim of the ball, which is never $\leq \kappa_0 Y$. The bound is attained, as follows.

Proposition 6 (Obstruction). *[PROVED; half-plane model] Take the dyadic boundary layer:*

- *row 0: unit squares on $y \in [0, 1]$, split by the forward diagonal;*
- *row $j \geq 1$: squares of side 2^j on $y \in [2^j - 1, 2^{j+1} - 1]$, each over two row- $(j-1)$ squares, cut into three triangles from the hanging bottom midpoint.*

This mesh has minimum angle $\arctan \frac{1}{2} = 26.57^\circ$, equal boundary edges, boundary stars $\cong \mathcal{S}_3$, and $\Theta \geq \sin 45^\circ$ at every vertex; the worst vertex is a coarse-square corner, with consecutive angles 90° and 45° . For every x_0 on the boundary and every $Y > 0$, $B(x_0, 3Y)$ contains a triangle of diameter $\geq Y$. So the hypothesis of Theorem 3 fails for every $\kappa < 1$.

Proof. • $Y < 1$. The boundary triangle at x_0 has diameter ≥ 1 .

- $Y \geq 1$. The point $x_0 + (0, 2Y)$ lies in row j with $2^j - 1 \leq 2Y < 2^{j+1} - 1$. Its triangle has diameter $\geq 2^j > (2Y + 1)/2 \geq Y$. $\square$

Placing this layer around the unit circle (polar columns, radial rows) is the evident construction. Its (M1), (M2) follow by openness, as in the penalty report. I have not written it out.

The witness still works on graded meshes [NUMERICAL] (`graded_check.py`, `graded_check3.py`). The boundary edge is 1, and "max κ " is the largest $\text{diam } T/Y$ over the support.

Y/h_Γ	1	1.5	2	3	4	8	16	32
dyadic (ratio 2)	12.36	14.82	15.28	15.44	15.46	15.485	15.488	15.488
max κ	1.41	1.49	1.12	0.75	1.12	1.12	1.12	1.12
uniform	12.36	14.82	15.28	15.44	15.47	15.486	15.487	15.487
triadic (ratio 3, 15.3°), $Y = 1, 2, 3, 9, 27$	12.36,	15.24,	15.42,	15.49,	15.48	(max κ up to 1.8)		

On the dyadic layer, $\mathcal{Q}$ tends to the continuum value *although* $\kappa \approx 1.1$ at every scale. The witness's quality is set by the fine cells near Γ_h . The coarse cells carry little energy, and the true interpolation error there is far below the worst-case bound. So on graded meshes the conclusion $\mu^* \gtrsim h/h_\Gamma$ appears to survive. Proving it this way needs a *sharp* interpolation bound on coarse cells, with $\kappa \sim 1$, which a Taylor/sup-norm argument cannot deliver. Two routes would close the general case:

- a computer-assisted, shape-uniform bound of the Argyris error functional for this specific F ;
- the half-plane-limit positivity of Theorem B in the penalty report, restricted to the explicit family $\{\text{curl } I_A F_Y\}$, which is compact under (M0).

Neither is done. [OPEN].

What is gained unconditionally. Together with Corollary 4:

- [PROVED] $\mu^* \geq 15 h/Y_*$, where $Y_* := \inf\{Y \leq Y_0 : \exists x_0 \text{ at an edge midpoint with all } T \cap B(x_0, 3Y) \neq \emptyset \text{ of } \text{diam} \leq \kappa_0 Y\}$. This is the " ρ_{loc} " statement: $\mu^* \geq 15\kappa_0 \rho_{\text{loc}}$ with $\rho_{\text{loc}} := h/(\kappa_0 Y_*)$.
- Since the boundary triangle has diameter $\geq |e| \geq c_0 h_\Gamma$, we always have $\kappa_0 Y_* \geq c_0 h_\Gamma$ and hence $\rho_{\text{loc}} \leq \rho/c_0$. Equality up to constants holds iff about $3/\kappa_0$ quasi-uniform layers exist somewhere along Γ_h .

6 3D (Goal 3) – not done

- *Theorem 3B (td_pen)*. The perturbation v_0 is controlled by β^{-1} from (IS3). Lemma IS3 gives β only through $C_{\text{loc}}(k, \sigma)$ of [GN18, Thm 3.1] and the Bogovskiĭ constant, neither of which is explicit. The 2D trick that avoids β is to use the curl of a C^1 interpolant. It would need an explicit-constant smooth de Rham interpolant on Alfeld splits: the FGN20 commuting projections, whose $W^{1,\infty}$ bounds are unverified and non-explicit.
- *td_k3b*. $\kappa_0(\theta, \theta_0)$ comes from compactness of the Star Nondegeneracy Theorem. An explicit version needs lower bounds for the rank-2 tangential moment away from the degenerate set $\mathbb{I} = \pm\nu \otimes \nu$. The report notes this is possible in principle.

Both are substantial and were not attempted here.

Files (all in notes/v6/consts/)

- `argyris_consts.py`, `.log`, `consts.json`: exact reference basis, Bernstein bounds, Hermite coefficients, profile norms, half-plane values (≈ 5 s).
- `assemble.py`, `.log`: $E_j(\theta)$, $\mathcal{Q}_{\text{lb}}$ and $\kappa_0(\theta)$.
- `validate_interp.py`, `.log`: physical Argyris check of Lemma 1.
- `realistic_kappa.py`, `graded_check.py`, `graded_check3.py` and their `.logs`: numerics on uniform, perturbed, dyadic and triadic meshes (under 2 min total).